RIPHAH INTERNATIONAL UNIVERSITY

Faculty of Computing

Department of Software Engineering

Mobile App Development

Semester Project: Student Portal App with Motivational Quotes

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The result of my semester project was a learning process and an understanding of mobile software through their building. I have taken my app to the next level by adding additional features that make it look completer and more professional.

1. Animated Splash Screen (Extra Feature)

One of the additional functionalities I implemented was a fully animated splash screen.

What I added

- A moving gradient background
- Floating emojis rising upward
- Smooth fade-in and fade-out transitions

Why I added it

The purpose of adding animation was to make the first impression of the app more appealing and professional. A simple static splash screen looks incomplete, so I enhanced it with modern visual effects.

What I learned

- How to use animation libraries in React Native/Expo
- How to manage timing, delays, and transitions
- How animated components affect user experience

2. Home Screen – Upcoming Events Section (Extra Feature)

What I added

- A complete "Upcoming Events" section
- A **View More** and **View Less** toggle button

Why I added it

To make the home screen look rich and complete, I added a dynamic events section. It creates a real-world feel, similar to professional apps.

What I learned

- How to show/hide components conditionally
- State management for dynamic UI updates
- Designing organized and user-friendly content sections

3. Interactive Calendar Component (Extra Feature)

What I added

- A full calendar view
- Highlighting of the current date
- Ability to browse dates visually

Why I added it

A calendar adds functionality, improves usability, and gives the app a complete look. It makes the home screen feel informative and functional.

What I learned

- How to integrate third-party components (calendar libraries)
- How to pass props for date selection
- How to style complex components according to the app theme

4. Animated Code Screen (Extra Feature)

What I added

- Custom animation on the code screen
- Movement/transition effects that enhance the UI

Why I added it

To avoid plain and boring layouts. Animation makes the screen visually engaging and keeps the user experience smooth.

What I learned

- Creating independent animations for different screens
- Performance considerations while animating multiple components

5. Complete Settings Page (Extra Feature)

This entire Settings screen was an extra addition done by me.

What I added

- Help & Support
- Contact Us

- Notifications
- Messages
- Profile and other general setting items

Why I added it

To ensure the app looks like a **complete, professional product** rather than a basic class assignment. A setting gives structure and accessibility to features.

What I learned

- Designing a side-menu/navigation layout
- Organizing features into categories
- Implementing clean UI/UX principles

6. Marquee State (Extra Feature)

What I added

A moving line at the top of the screen that continuously scrolls text horizontally, similar to professional apps and websites.

Why I added it

It makes the interface look lively, updated, and modern. It also fills empty space and enhances the overall appearance of the app.

What I learned

- How to implement continuous looping animations
- Designing attention-grabbing but non-intrusive UI elements

Issue Faced:

During the development of my mobile application, the major issue I faced was related to CSV file generation. When I tried to generate the CSV file, the data was not fetching correctly from the database, which caused the CSV file to be empty, incomplete, or incorrectly formatted. Sometimes the CSV file was generated, but the values inside were arranged incorrectly or missing due to asynchronous data loading delays.

The issue mainly occurred because:

- The data was not fully loaded before the CSV export function was executed.
- There were asynchronous state updates.
- The mapping of the dataset to CSV format was not structured correctly.
- The CSV library required a specific format, but the data I passed did not match that format.

This caused the CSV export to fail or produce incorrect files.

Solution:

To solve the issue, I used a more structured and reliable CSV generation method with proper data formatting and asynchronous handling.

Step 1: Ensuring Data Was Fully Loaded

This was done by:

- Awaiting the API call or Firebase query
- Verifying that the data array was not empty
- Logging the data to confirm that all fields were properly received

I made sure that the data was completely fetched before generating the CSV.

Step 2: Converting Data into a Proper CSV-Friendly Format

```
const formattedData = data.map(item => ({
  name: item.name,
  email: item.email,
  phone: item.phone,
  // all other fields...
```

```
}));
```

CSV files require data in a consistent structure (array of objects with matching keys).

Step 3: Using a Stable CSV Generation Method

```
import { unparse } from 'papaparse';

const csvString = unparse(formattedData);
```

I implemented a reliable CSV generation library such as papaparse.

Step 4: Creating & Downloading/Saving the CSV File

```
import * as FileSystem from 'expo-file-system';

await FileSystem.writeAsStringAsync(
  FileSystem.documentDirectory + "students.csv",
  csvString,
  { encoding: FileSystem.EncodingType.UTF8 }
);
```

I used file-system functions to save the file, ensuring no missing data or formatting issues.

Apply themes to Text, Inputs and Buttons:

```
<Text style={{ color: theme.mainText }}>Some Text</Text>

<TextInput
  style={{ backgroundColor: theme.inputBg, color: theme.mainText }}
  placeholderTextColor={theme.placeholder}
/>

<TouchableOpacity style={{ backgroundColor: theme.button }}>
  <Text style={{ color: "#fff" }}>Button</Text>
</TouchableOpacity>
```

Feedback:

New Things I Learned:

- Animated splash screens
- Gradients and floating emoji animations
- View More/View Less expandable sections
- Calendar integration
- Setting Screen (Help, Contact, Notifications, Messages)
- Marquee State
- Third-party libraries
- CSV file generation
- Debugging asynchronous issues
- UI/UX improvements

Overall Experience:

This project improved my problem-solving, UI/UX skills, debugging abilities, and overall understanding of mobile development. The extra functionalities made my app look like a real-world application rather than a basic academic assignment.